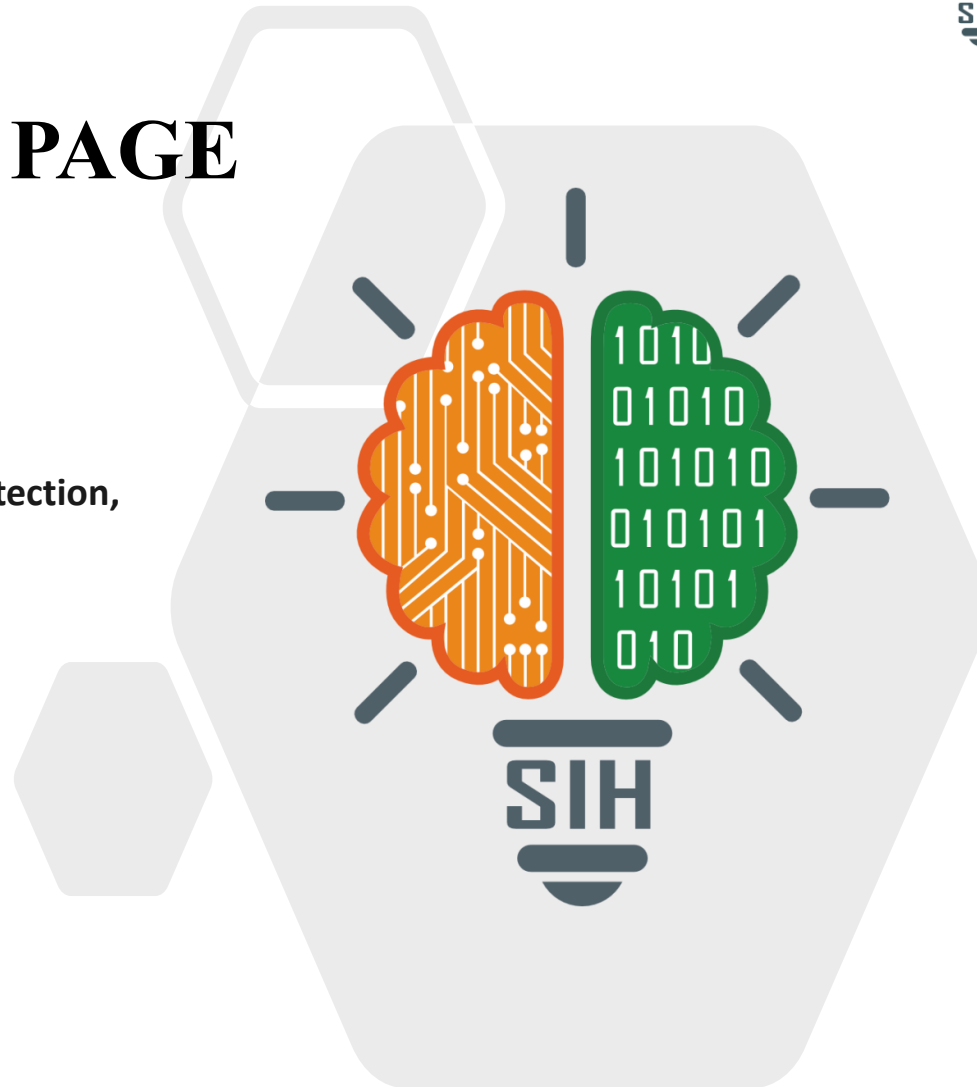


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26106
- Problem Statement Title – AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform
- Theme – Blockchain & Cybersecurity
- PS Category – Software
- Team ID – [Fill from SIH portal]
- Team Name (Registered on portal) – prajjwaldubey





Proposed Solution:

- Develop a comprehensive web platform that analyses original emails (paste or .eml upload), focusing on improving institutional email security through technology.
- Features include SPF/DKIM/DMARC checks, scam-language cues, lookalike domain spotting, hop-path geolocation, explainable risk score, and a downloadable forensic PDF.

Detection, Trace & Forensics:

• Threat Scoring:

Integration of authentication stamps with content cues; identifies trends and labels legitimate / suspicious / impersonated / phishing / fraud.

• Path Mapping:

Rebuilds the Received-header chain and plots public IPs with city, country, and ISP.

• Case Vault:

Digital case store with file fingerprint, minimising rework and promoting shared review across SOC/admin teams.

How it Addresses the Problem:

• Detection Engine:

Spots phishing, BEC, and spoofed mail using headers + body patterns, helping institutes catch fraud before users click.

• GeoLocation Module:

Shows where mail servers sat along the path — useful for investigation without naming a person's home address.

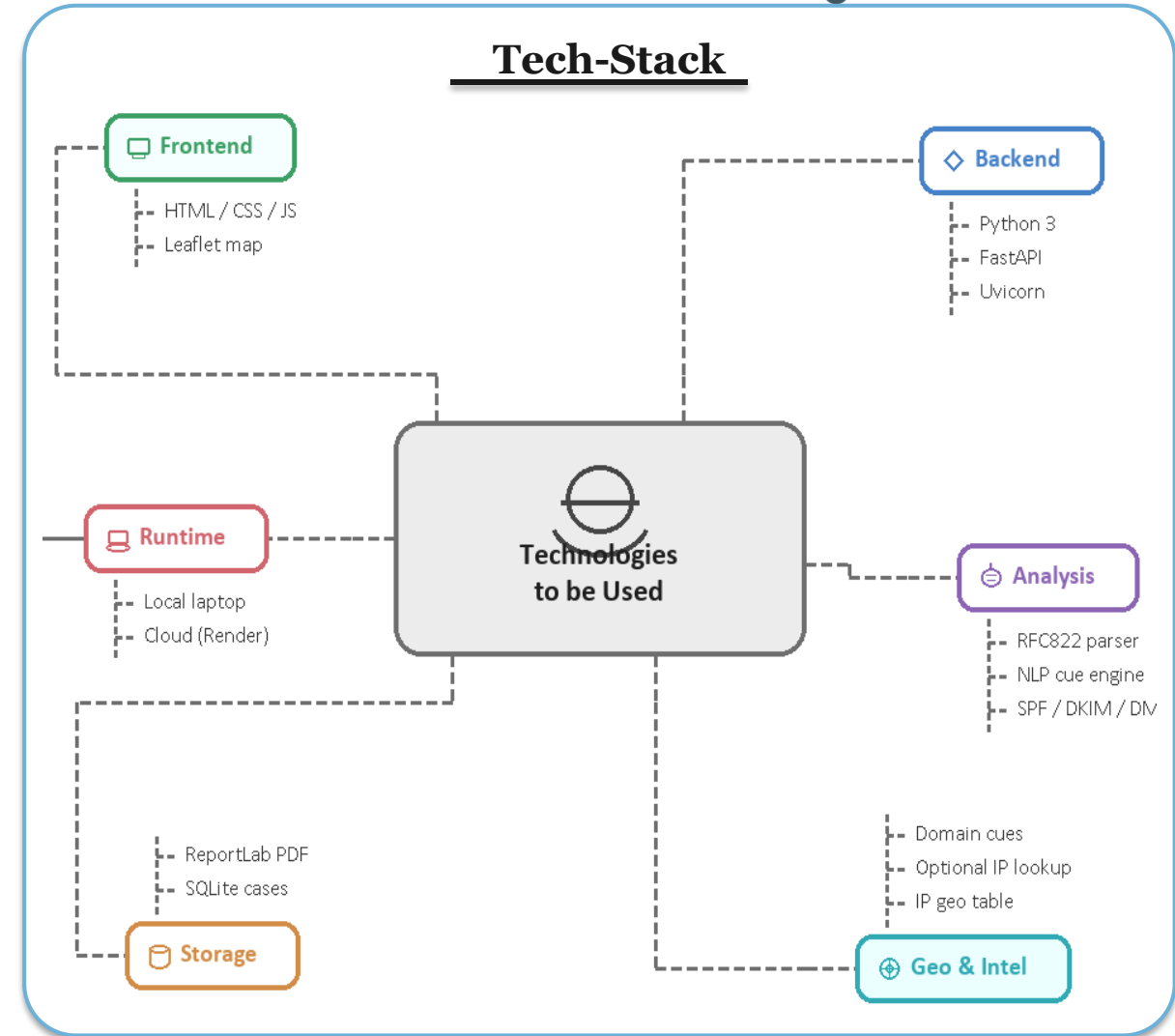
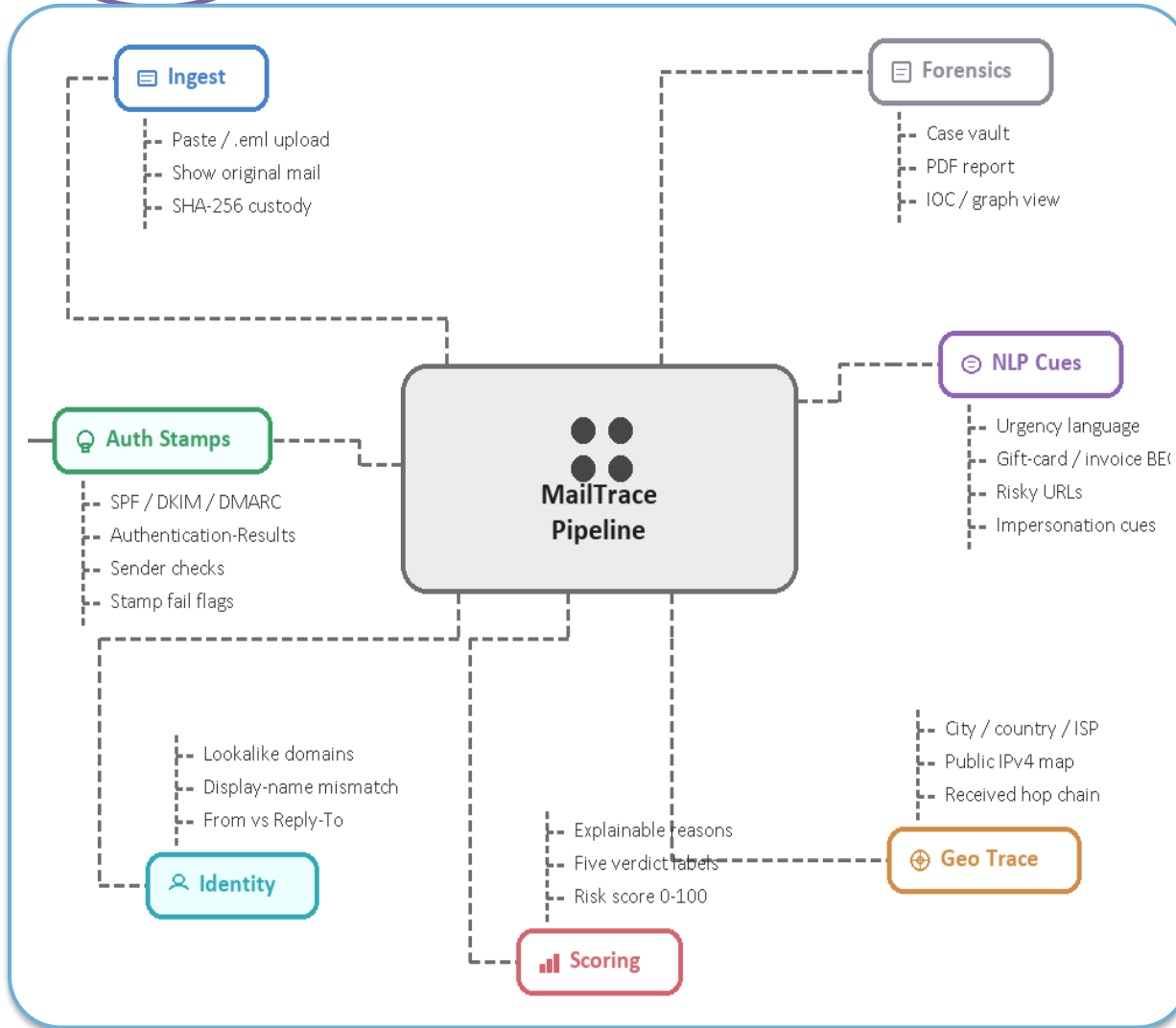
• Forensic Report:

Ready PDF with score, reasons, stamps, and map for tickets and incident response.

Innovation and Uniqueness:

- End-to-end flow: detection + map + PDF in one laptop tool.
- Explainable score — every point has a written reason.
- Honest geo: mail servers / ISP only, not personal identity.
- Works on real Gmail “Show original” and .eml files.
- Graph of domains, IPs, and clues for campaign-style review.
- No mailbox login; user pastes the mail they already received.
- Chain-of-custody fingerprint for investigative integrity.

TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

Feasibility

Software & Cloud

- Laptop demo ready
- No RF hardware
- No classified data

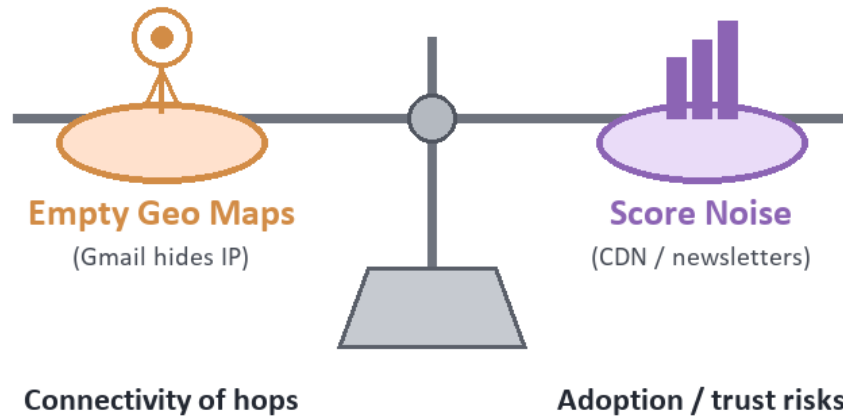


Feasibility Analysis of
MailTrace Platform

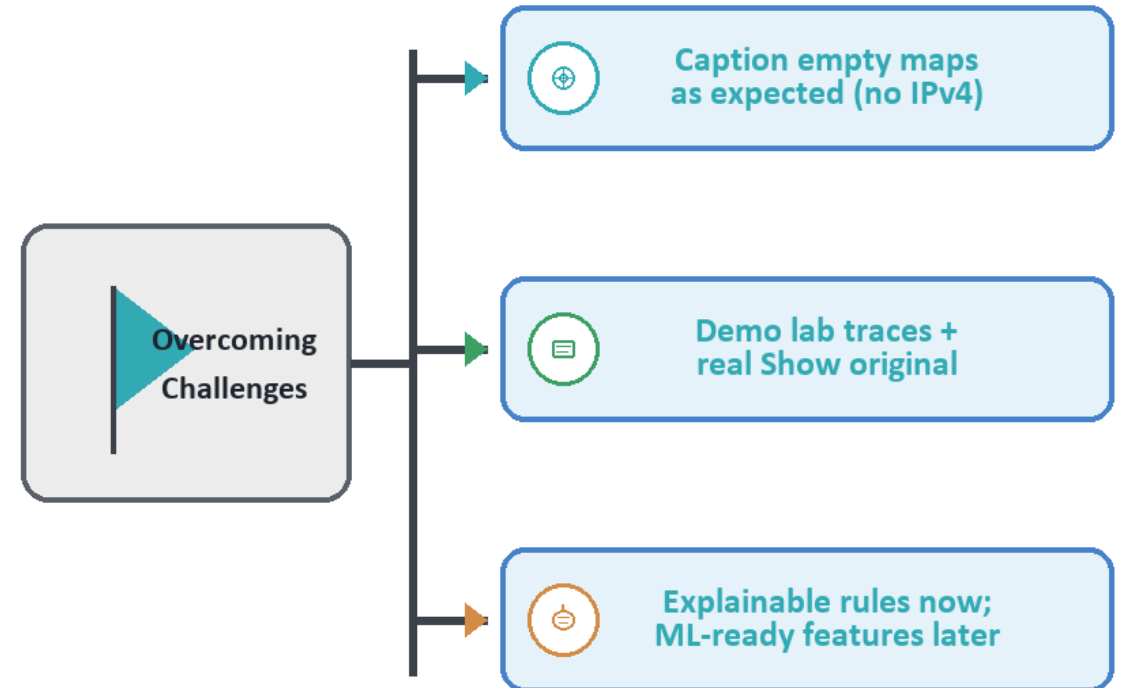
Current Technology

- Header forensics
- NLP scam cues
- Hop geolocation

Challenges and Risks



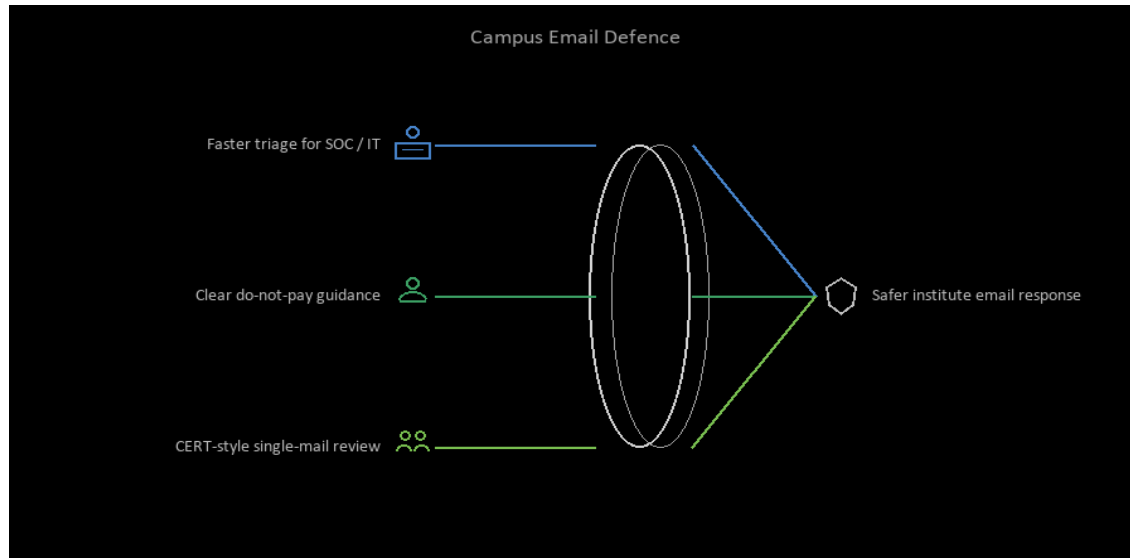
Overcoming Challenges



IMPACT AND BENEFITS

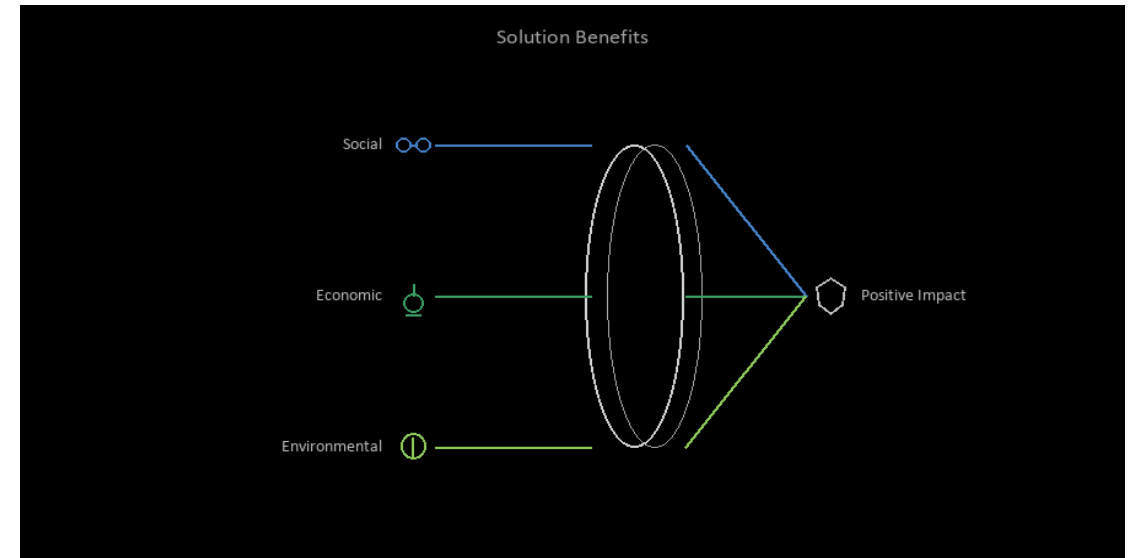
Potential Impact on the Target Audience:

- Improved threat awareness and faster triage of fake dean, GST, invoice, and internship-scam mail for college / institute SOC and IT admin.
- Empowerment of students and staff with a clear “do not pay / click” vs “looks legitimate” view.
- Enhanced readiness for CERT-style review of a single suspicious message without reading raw headers by hand.



Benefits of the Solution:

- Social: Bridges the gap between spam filters and real investigation; promotes safer campus email use.
- Economic: Cuts fraud loss and analyst time per ticket; PDF supports internal records.
- Environmental: Reduces paper usage with digital forensic reports.



RESEARCH AND REFERENCES



Email Authentication Standards

- **RFC 7208 — SPF**
How receivers check if a host may send for a domain.
<https://datatracker.ietf.org/doc/html/rfc7208>
- **RFC 6376 — DKIM**
Cryptographic signature so body/headers can be verified.
<https://datatracker.ietf.org/doc/html/rfc6376>
- **RFC 7489 — DMARC**
Policy aligning SPF/DKIM with the visible From domain.
<https://datatracker.ietf.org/doc/html/rfc7489>

Threat Landscape and Practice

- **CERT-In / MeitY advisories**
Guidance on phishing and business email compromise in India.
<https://www.cert-in.org.in/>
- **APWG Phishing Activity Trends**
Shows volume of credential and payment fraud mail.
<https://apwg.org/>
- **RFC 5321 — SMTP**
Mail is relayed; Received chain is servers, not a person's house.
<https://datatracker.ietf.org/doc/html/rfc5321>